TO STUDY COST PREDICTION ANALYSIS OF CONSTRUCTION PROJECT USING ANN MODEL (ARTIFICIAL NEURAL NETWORK) and SVM BY MATLAB

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Abstract- Cost estimation is an experience-based task, which involves evaluations of unknown circumstances and complex relationships of cost-influencing factors. An artificial neural network (ANN) is an analogy-based process, which best suits the cost forecasting domain. The primary advantages of ANNs include their ability to learn by examples (past projects), and to generalize solutions for forthcoming applications (future projects). Construction cost prediction is important for construction firms to compete and grow in the industry. Accurate construction cost prediction in the early stage of project is important for project feasibility studies and successful completion. There are many factors that affect the cost prediction. This research presents the comparison between ANN and SVM. The objective of this paper is to develop neural networks and multilayer perceptron based model for construction cost prediction. Estimating construction costs and predicting price escalation are major steps for project owners, estimators, and contractors. Therefore, the cost estimation plays a significant role in construction project decisions and represents the most important corner in iron triangle of construction management. This project is implemented on actual site case study located at Ambegaon Pune under the management of SP Construction Pune (Ganesh Construction).In order to success theconstruction projects we need technique to estimate the cost with high degree of accuracy and less error. In the present investigation, the model built by applying both quantitative approach and qualitative approach to identify the factors (variables) as inputs of the model. 85 projects are used for developing, training and testing the ANNs model, the output of this model is the expected construction costs of the projects. To validate the model, 14 projects as sample have tested to predict the cost with high degree accuracy and acceptable error. Consumer Price Index, cost of construction materials, type of building, market conditions, structural system, site Area, type of slab, other Supplementary buildings, location of the Project, project Size, type of foundation, building closeness, and fluctuation in the Currency are the main factors affecting in construction buildings costs. These factors have been used as inputs in ANN model and all data is extracted from the historical projects, the model has been developed and trained for 70 projects and compared the actual cost with predicted cost. The model was validated throughout sample of projects. 13- 17- 1 model was the best between 15 models are developed, 6% is the mean absolute percentage error for model is tested. The results are clearly provided a good indicator for predicting the construction buildings costs in the future with high degree of accuracy.

Keywords: Cost Factors, Artificial Neural Network System, Feed Forward Network,Multilayer Perceptron, Back Propagation Error, Developing the model.

1. INTRODUCTION

Neural computation is one of the inductive machines learning methodologies, it is most often used to learn, generalize and represent general knowledge. It extracts information from existing data by inductive learning. It is a fundamentally different approach to other information processing approaches. Algorithmic computing is used in cases where the processing can be described as a known procedure or a set of known rules. Neural computation allows the development of information processing for which the rules and relationships knowledge are not available (Hecht-Nielson 1990). Expert systems require rules or instructions, which are executed one at a time to arrive at an answer. By contrast, artificial neural networks take in a great amount of information at once, and then draw a conclusion. Once taught, an ANN looks at new input data and produces an answer instantaneously.

1.1 General

A back-propagation neural network consists of a number of layers; each layer synthesized of different number of neurons (processing elements) as depicted in Figure 2. The input layer represents influencing factors of a specific problem. The output layer in which the solution of the problem takes place, e.g. prediction, classification, etc. The hidden layer through which the information is processed. The numbers of hidden layers and hidden neurons are usually determined by trial and error according to the complexity of the problem.

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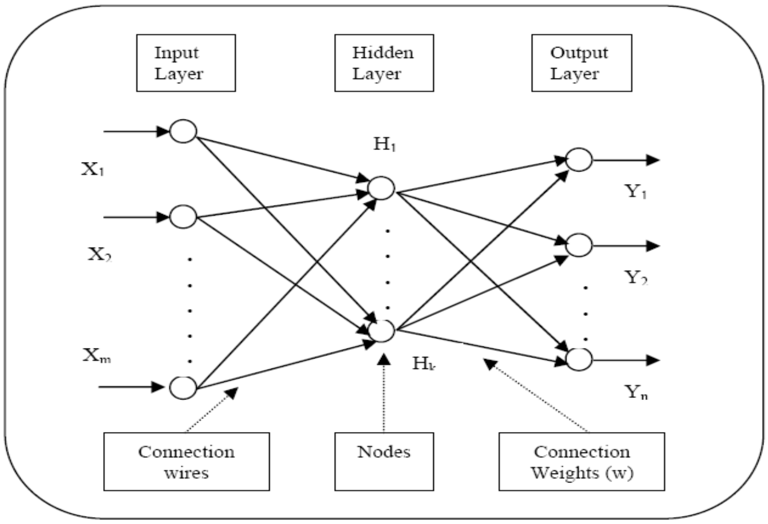

Fig01. A Simple Artificial Neural Network Structure.

Project cost management is all about controlling cost of the resources needed to complete project activities. Apart from these controllable costs, there are certain aspects over which we do not have any control. These are called uncontrollable costs and they are subject matter of risk management, taken up separately. Figure 1 is illustrated bellow three corners project constraints in construction project management, the cost, time, and quality are callediron triangle of the project. Artificial neural networks try to reproduce the generalization abilities of a human neural system. Neural Networks are particularly effective for complex estimating problems where the relationship between the variables cannot be expressed by a simple mathematical relationship.

1.2 Artificial Neural Networks used globally

Artificial Neural Networks are the computational models that are inspired by the human brain worldwide. Many of the recent advancements have been made in the field of Artificial Intelligence, including Voice Recognition, Image Recognition, Robotics using Artificial Neural Networks worldwide. Artificial Neural Networks are the biologically inspired simulations performed on the computer to perform certain specific tasks like –

1. Clustering
2. Classification
3. Pattern Recognition

Artificial Neural Networks, in general – is a biologically inspired network of artificial neurons configured to perform specific tasks. These biological methods of computing are considered to be the next major advancement in the Computing Industry. They are computer programs simulating the biological structure of the human brain which consists of tens of thousands of highly interconnected computing units called neurons. An Artificial Neural Network can be constructed to simulate the action of a human expert in a complicated decision situation. Construction management and cost engineering have made tremendous advances to address overrun and delay, minimizing uncertainties, productivity estimation. A neural network is applied for prediction performance construction projects, estimating LCC cost at each stage of construction projects; training models without need for more detailed design, ANN approach provided good predicting ability despite the no uniform distribution and incompleteness data
[1-3]. A few techniques such as regression, and traditional methods, artificial neural network technique is utilized for predicting costs in various construction projects, which address the problems to enable the estimator of cost predicting accurate cost in construction building projects.

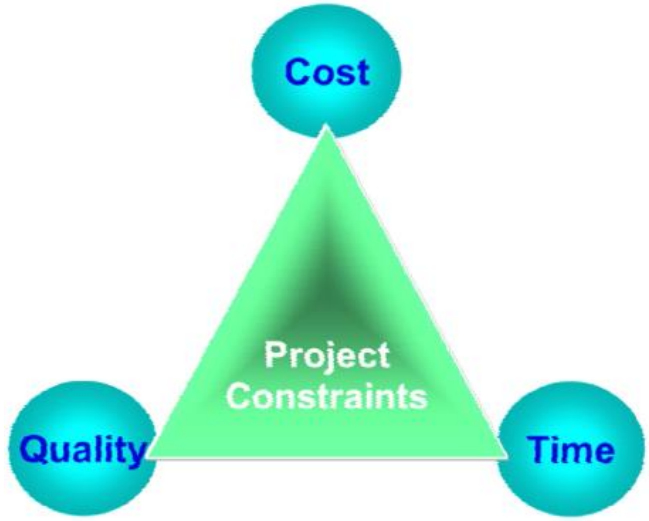

Fig. 02: Iron Triangle in Construction Management

Project owners and construction project managers need a way to focus and prioritize their efforts to control project costs, when their efforts can have maximum impact on the total cost of the project. In the earliest phase of planning and design, only the most basic and functional decisions about the project have been made and the data available for predicting project costs is ambiguous and highly subject to change. Under these conditions, traditional methods for predicting the cost such as built-up, unit cost and expert system estimating become inaccurate or impossible to implement. Stakeholders responsible for controlling project costs are in need for an alternative technique to the traditional cost-based prediction methods to help them predict the cost of their projects using the limited available data in the early phase of the project. The limitations identified with the conventional models stimulated the researchers not to rest on their oars, the drive to evolve better method then became the order of the day. The existing models then were not challenged since they lack applicability until advocators stressed the need to depart from existing research status and go for research output that can be backed with solid theory. He doubts the reliability of existing forecasting models and urged the development of good forecasting method with solid framework for applicability. Artificial neural networks are nonlinear information (Signal) processing devices, which are built from interconnected elementary processing devices called neurons. An Artificial Neural Network (ANN) is an information-processing paradigm that is inspired by the way biological nervous systems, such as the brain, process information. The key element of this paradigm is the novel structure of the information processing system.

1.3Problem Statement

In conventional approach Cost is an important aspect to everyone, especially in the construction projects. For any project requires accurate cost prediction in order to inspire the decision either forward or cancel the project. Moreover, predicting the cost plays a key role in the successful completion of the construction projects. Due the lack in information, details, drawings1.5 and many important factors that affecting in estimation the cost during planning phase, the project will be at risk. Therefore, the cost estimation plays a significant role in construction project decisions and represents the most important corner in iron triangle of construction management.

1.4 Objectives of the Project

1) To study and identify main factors affecting in construction buildings costs.
2) To use a new technique to estimate the cost with high degree of accuracy and less error.
3) To develop and train the ANN model using MATLAB software to build by applying both quantitative approach and qualitative approach to identify the factors (variables) as inputs of the model.
4) To analysis the factors affecting cost of a construction project.

1.5 Scope of the project

The scope of this project is to develop a new technique to estimate the cost with high degree of accuracy and less error and to develop and train the model using Matlab to build by applying both quantitative approach and qualitative approach to identify the factors (variables) as inputs of the model. This project is implemented on actual site case study located at Ambegaon Pune under the management of SP Construction Pune (Ganesh Construction)

2. LITERATURE REVIEW

Construction costs are among issues that are highly pronounced in construction practice and their investigation has become a challenge for many authors worldwide. They have investigated different aspects of construction costs such as: costs estimation, costs forecasting, costs overruns, the relation cost-time. For estimating and forecasting the construction costs there are traditional models which are based on quantities, design, resources, functional elements or building operational units etc. But, many authors put particular attention on the untraditional models for costs estimation and forecasting. They use new techniques and practices such as the experimental models, regression models and simulation models [1].

Authors in have tested and have refitted linear regression model. They have shown that various projects need various estimates of parameters. They developed two type of models: for industrial projects and for projects that are not industrial. Similarly, other authors investigated the correlation between time and cost overruns by applying the time-cost algorithm [2].

The functional relationship construction time -construction cost for highways was explored. Additionally, for forecasting cost and time regression models were created. Models depend on contract sum, contract period, contractual arrangement, knowledge of the contractor selection method, client sector, project type etc. Regarding sensitivity analyses it was shown that the errors in actual construction cost forecasting for large and small projects are virtually the same. In proposed a probabilistic model for predicting risk effects on projects costs and duration. For developing the statistical regression model and sample tests they used historical data of similar construction projects. With the 95% probability of the model the precision of the mean cost and time prediction was ±0.035% [3].

A paramedic model for estimation of cost for highway projects was presented. A supervised neural network model optimized with Genetic Algorithms was established together with parameters of determination which notably impact the price of the highway projects. The results showed that the developed model is reliable to be used at early stages of highway projects because its percentage error is 16% (and is lower than the allowed 20%). Similarly, gives cost estimation model that is based on artificial neural network and is useful for highway projects at the conceptual phase in developing countries [4].

A model for cost estimation in road projects that is useful for construction managers is built. The model is developed using support vector machine. The 12 factors were identified to be the most important factors affecting the cost-estimating model. A total of 70 case studies from historical data were used for modelling and the built model was successfully able to predict project cost with accuracy of 95%. A parametric model for estimation of cost for railway systems in urban areas in predesign stage. Incorporating neural networks and regression analysis, the authors have developed powerful parametric model for selecting the parameters with big impact on the cost during early project stages [5].

Predictive regression model that uses factors which affect project cost and examined their importance. From the literature review in this study it is indicated that significant effects on the overall project cost come from several factors. They are connected with: clients, expert's duration, etc. Some of the six most important factors to project cost identified by the study are the levels of: design complexity, construction complexity; technological advancement etc. The contractor's experience on similar projects; the consultant and the client; the suitableness of contractor's plant and equipment used are the most important among those factors, so they were used for developing the cost predictive model [6].

For estimation of the construction cost the back-propagation network (BPN) model is applied. General algorithms are incorporated in BPN in order to choose the parameters of BPN's. The authors obtained very effective and accurate model for estimating construction costs. An efficient cost estimation tool is presented, useful for project managers and designers in early phase of design process for buildings, using neural network methodology for estimating the cost of square meter for 4-8 storey residential reinforced concrete-structural systems. The accuracy that was achieved was 93% [7].

In support vector machine, an artificial intelligence technique is used for construction cost estimation that is useful for planners and owners for predicting the cost of a construction project. Through interview with experts and literature review, the factors which impact the cost estimate most, are identified. The data from 29 construction projects are used in the training process of the SVM. The average prediction error of the model that they obtained was less than 10% and the computation time was less than 5 minutes [8].

Traditional methods of estimating project costs do not attempt to assess the magnitude of the variation inherent in the estimate. As a result, there is a risk that decisions on strategy selection will be based on a high degree of uncertainty. The traditional approach to cost estimating is to derive a best estimate from a knowledge of existing conditions based on current rates and prices in similar situations with adjustments to reflect anticipated differences in ground conditions, site accessibility, and other factors [9].

Several estimation methods are used in construction practice and the suitability of any particular method is usually dependent on the purpose it is used for, the amount of information available at the time of estimation, and the party using it. Despite the reliance of clients and contractors on available cost estimation and forecasting methods, the actual final costs of construction projects still considerably deviate from their original estimates [10].

To propose a generic copula-based Monte Carlo simulation method for prediction of construction projects' total costs with dependent cost items. Found that different dependence structures can lead to different probability distributions of total cost. Also finds that the existing goodness of fit tests can be employed in choosing the best performing copula. The paper concluded that the copula-based Monte Carlo simulation method can predict total cost of construction projects with reasonable accuracy [11].

A novel approach from the domain of forecasting. The multistep ahead (MSA) approach relies on the idea of using several cascaded estimations to predict future values. Accordingly, building element quantities were estimated as the first step. In the second step, estimated quantities were combined with the existing set of inputs to achieve a higher accuracy in construction cost prediction. In order to test the hypotheses of interest, 657 building projects from Germany were analyzed using linear regression and artificial neural network methods. Conclusive evidence suggests that the MSA approach significantly outperforms the prediction accuracy of the conventional practice [12].

Hybrid Artificial Intelligence on Schematic Design Stage: RANFIS and CBR-GA and compared the performance of cost estimation models of two different hybrid artificial intelligence approaches: regression analysis-adaptive neuron fuzzy inference system (RANFIS) and case based reasoning-genetic algorithm (CBR- GA) techniques. Models were developed based on the same 50 low-cost apartment project datasets in Indonesia. Tested on another five testing data, the models were proven to perform very well in term of accuracy. A CBR-GA model was found to be the best performer but suffered from disadvantage of needing 15 cost drivers if compared to only 4 cost drivers required by RANFIS for on-par performance [13].

Research on Forecasting the Cost of Residential Construction Based on PCA and LS-SVM. He forecasted the costs of a residential construction rapidly and accurately in the initial stage of construction and lack of relevant information. Based on the strengths and weaknesses of previous studies about it, a new model to forecast the costs of a residential construction which is based on Principal Component Analysis (PCA) and Least Squares Support Vector Machine (LS-SVM) is proposed. Finally, selected 5 projects in conjunction with the new model for simulation analysis, the relative error are controlled within±7% [14].

Resolving it's Challenge to Statistical Modeling and he used a Bayesian approach for combining the reference class forecast and the model-based forecast. He used this approach to estimate healthcare costs under a Voluntary Employee Benefit Association (VEBA). The accuracy of three estimating techniques (regression analysis (RA), neural network (NN), and support vector machine techniques (SVM)) by performing estimations of construction costs. By comparing the accuracy of these techniques using historical cost data, it was found that NN model showed more accurate estimation results than the RA and SVM models. Consequently, it is determined that NN model is most suitable for estimating the cost of building projects [15].

3. METHODOLOGY

3.1 .Methodology:

Fig. 03:Flow chart of project.

3.2. Methodology

To develop the parameters cost estimation model by applying the artificial neural networks methodology is the essence of this research, which takes the building projects to predict the final cost of construction building projects. This study adopts upon the historical data to training this methodology, using the main factors affecting as inputs to provide historical data to make estimation cost for new projects. The necessary information and required projects data were collected to build the model of estimation construction cost of building projects. Conceptual cost estimation, cost control, cost overrun, cost codes and cost management are reviewed to identify the main topics to be handled in this research. All types of artificial neural networks (ANNs), their structure, and applications are outlined. To achieve the objective of the research, development artificial neural networks model for construction projects cost. The research strategy can be classified into two types namely, quantitative approach and qualitative approach. The both types quantitative and qualitative approach are used to collect thee data of the projects and thee information about the factors which influencing in the construction building cost.

This research methodology consists from two stages:

3.3. Research Strategy:

3.3.1) Theoretical study: It is involving review the cost estimation and (ANNs) technique from the references, thesis, journal papers, and books were published in project management field relating to the subject of research

3.3.2) Field work:It is involving the following steps:

3.3.3) Description and selection the factors that influencing the costs of building construction projects.

3.3.4) Pilot study involving interviews are used to modify and improve the draft questionnaire

3.3.5) Factors are identified and categorized in different groups before sending the final questionnaire to collect the respondents

3.3.6) A questionnaire survey methodology is employed to determine and rank these factors according to their levels of influence onthe cost of building construction project using SPSS program.

3.3.7) Developing of NN model to predict the cost of building construction projects using MATLAB SOFTWARE program.

3.3.8) Validation of the NN model

3.4. Historical Data Collection

Based on thesis methodology and after determining the most important factors, data for training and testing the proposed artificial neural networks model were collected. These data were gathered from real life projects conducted from 2000 to 2017 in Yemen. This process was done using a data collection. (86) Construction buildings projects are collected from Yemen. All data required to construct the model have been collected from ministry of work in Yemen and the companies that executed these projects to compare the estimation value with actual cost.

3.5. Cost Prediction Using Neural Networks:

This chapter explains the incorporation of the previously identified cost factors into construction buildings. These factors that affecting in construction buildings costs are investigated to be as inputs in ANN model to predict the total cost of construction buildings in Yemen. As mentioned in chapter three, 85 projects are prepared and arranged for developing the ANN model. The data is deduced from 85 projects by Microsoft Excel to import it directly into MATLAP program as inputs and outputs. First step in this chapter is designing the ANN model regarding on the data was collected and analyzed, thereafter, throughout the artificial neural networks technique is applied to prepare cost prediction model. Finally, three processes have done by MATLAB program namely, training, testing, and validating the best model with lees error and highest accuracy.

3.6 Design the ANN model

Designing ANN models are divided into five basics steps namely, collecting the data, preprocessing data, architecture the network, training the model, and testing the performance of model.

1) Data Collection: Collecting and preparing the data is the first step in designing the model. As it is outlined in previous chapters, the main factors that affecting in construction buildings costs and the historical projects (total cost) in period from 2000 to 2) Data Pre-processing : After data collection, the variables were analyzed using mathematical and software methods such a Microsoft Excel and SPSS23 program, to select the main factors that affecting in construction buildings costs. The data preprocessing procedures are conducted to train the ANNs model more efficiently.

These procedures are,

2) Solve the problem of missing data from the projects, (2) normalize data, and (3) randomize data. The missing data is replaced by Assuming the value that is the nearest to reality such as, some projects the area of them is missing and not founded, regarding to the insulation layer of building surface in (m2) is taken as area value.

3) Architecture the Network: The neural network was built to consist of three parts, the input layer, hidden layers, and output layer. The weights are linked between these layers, all nodes enclosed by weights from inputs to hidden layer and also from hidden layer to output layer.

4) Input layer, the first one, enclosing 13 nodes, each node represents one of the selected cost indicators, as concluded in the previous chapter, each cost indicator will be determined by a value; for quantitative indicators, the exact value will be used, however, for qualitative indicators, a preselected value will be used as an indicator for each case, table 1 illustrates the cost indictors used in the input layer and the correspondent determination value for each cost indicator. Finally, all the data in this layer will be scaled from (-1) to (1).

5) Hidden Layer(s), the second ordered layer. In this layer(s), the number on nodes (hidden nodes) ware calculated by considering one guidance that the number of hidden nodes must be not less than half the summation of the number of nodes in the input and output layers (Hosny 2011).

Accurate estimation of quantities and costs is a crucial factor in success of construction projects. The main complexity in the cost prediction is dependency on several factors. From those factors one of the most important factors is materials cost. The material cost is main impact factor in construction project for cost estimation which is varying every year. Cost of cement, sand, steel, aggregates, mason, skilled worker, non-skilled work workers the major factors which are considered in cost estimation. As shown in Table.1 six input and one output variables are extracted from the collected dataset [2] in which data of last twenty-three years has been collected from Schedule of rate book (SOR) and general studies.

3.7. Data Collection Site

For implementing this idea author used below site for implementation and analysis-

1. Name of the project : Ganesh Construction
2. Location : Ambegaon Pune.
3. Owner : SP Construction Pune
4. Purpose of project: Industrial purpose./ Commercial
5. Consultant: PankajMarathe.
6. Total Area of construction : 2,90,000 sq.ft.
7. Cost of the project : 25.4 crore.
8. Description- G+5

3.8 Support vector machines (SVM) predictive model

The Support vector machine (SVM) is a relatively new modelling method that has been rapidly developed in recent years, showing very good results at obtaining accurate models for many problems like pattern classification, function approximation, and regression problems. The architecture of SVM models is very similar to that of neural networks. There are two categories of support vector machine models, depending on the kind of the problem they solve: support vector regression (SVR) and support vector classification (SVC).Till now SVM research has been intensively oriented toward solving practical problems, evolving in very active field of investigation. Because of its very good generalization to unknown data, SVMs have demonstrated very good performances at prediction with time series and also regression problems and in the last several years SVM methods are becoming standard methods of machine learning. SVM are also called kernel methods, i.e. methods which use kernel functions to transform non-linear learning problem in linear by transforming the input space into output multidimensional space with kernel mapping functions and solving the problem in the output multidimensional space. This holds for solving regression problems and also classification problems. Fig. 2 shows mapping from two-dimensional into three-dimensional space [4]. This concept of mapping the input space into new multidimensional space and solving the problems in the new space is very efficient, allowing SVM models to perform very complex separations.

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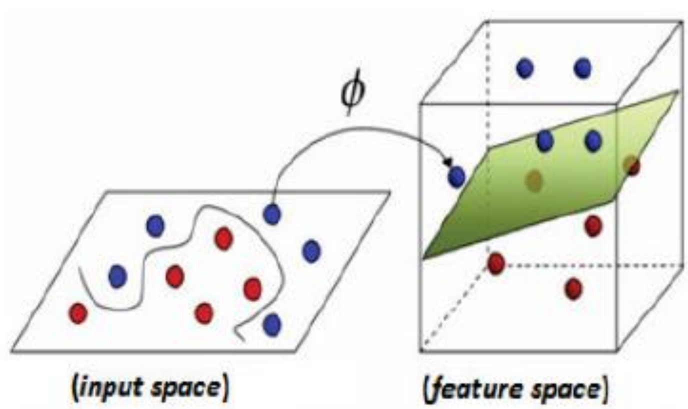

Fig.3.1 Mapping in multidimensional space

The novel training concept implemented in SVM shall be explained shortly. Similarly to neural networks SVMs have the ability to approximate arbitrary multidimensional function to some desired level of accuracy, and so they are very desirable for modelling non-linear and unknown processes. The only information available for learning is the input data set S

$$S = \{(x_i, y_i) \in X \times Y\}, i = 1,2, \dots \dots, m \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots (6)$$

Which contains m data pairs from which SVM will be trained to learn the relationship between input predictors and output target variable, obtaining some approximation function f(x,w). Are the input n-dimensional vectors, and are the values of the target variable, or outputs of the SVM model. W is weight vector whose components will be obtained in the process of training [10]. ixX∈iyY∈

After the process of training support vector regression (SVR) model obtains the non-linear n-dimensional approximation function f(x,w) which has some deviation from the actual target values from the training data set, which is, in fact, the error of the approximation. The accuracy of the SVR is being measured by computing the error of the approximation. The Vapnik's-insensitivity error (or loss) function is widely used.

$$L_\varepsilon = |y - f(x, w)|_\varepsilon = \{\frac{0}{|y - f(x, w)| - \varepsilon} \text{ For } |y - f(x, w)| \leq \varepsilon \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots (7)$$

E is the maximum distance of the function f from the actual values of the target variable, i.e. the values of the function f must lie in a tube with radius. Solving the SVR problem will be shortly explained on a linear regression problem. In this case, the approximation function f that should be obtained is linear, and it should approximate all data pairs (xi,yi) with maximal deviation ε (Eq. (8) and Fig.4):

$f(x, w) = w^T x + a$…………………………………… (8)

Because the width of the tube is inversely proportional to the norm of vector w, , the maximal allowed distance of the pairs (xi,yj) will be computed by minimization of . If for computational convenience (without changing the result) this condition is defined to be minimization of instead of minimization of and considering Eq. (7) and Eq. (8), the solution of this problem is reduced to convex optimization problem given by the conditions (9):

$\frac{1}{2}||w||^2$………………………………………………………………..(9)

In most of the cases the regression problems are non-linear. In this case, SVR algorithm solves the problem by mapping the input n-dimensional space (defined by input n-dimensional vectors xi) into some multi-dimensional, so called feature space with some mapping function Φ, called kernel function. The role of the kernel function is to transform the non-linear problem from the input space into linear problem in the new feature space. Then the problem in the new feature space is solved as linear regression problem, as was discussed above [5].

The parameter which controls the stopping of the optimization process of SVM algorithm is called Epsilon value, which is tolerance factor whose value can be changed by the user. By its reducing, a more accurate model will be generated, and by its increasing the computation time will be reduced. There are different methods for validating the accuracy of the SVM model in the DTREG software. They are: 1) random percent validation when random percent rows from the input data set are held out and used for validating of the model after the training process; 2) using variable for choosing the rows for validating; and 3) V-fold cross validation, when V-1/V proportion of the rows from the data set is being used in each SVM model during the training process..

4.2Predicting construction cost using ANN Model

4. 2.1 Data Collection and Identification of Input Variables

Structural data from a total of thirty (30) building projects were collected from different construction companies in the Metro Manila. The structural costs obtained were for bidding purposes. Six variables namely: number of stores, number of basements, floor area, volume of concrete, area of formworks, and weight of reinforcing steel were considered. The number of storeys ranges from 5 – 37 floors while the number of basements is between 0 – 3. The total structural cost of buildings ranges between PHP 2-448 million. There were missing values in the data provided but as mentioned in literature, ANN is capable of analyzing incomplete data sets. Also, the period of data collection as well as the time at which the cost estimation of the buildings was done, was not taken into consideration in this study. Doing so could have improved the results as inflation rates affect the cost of materials. Arafa and Alqedra (2011) considered seven (7) factors affecting the structural system cost, namely: ground floor area, typical floor area, number of stores, number of columns, type of footings, number of elevators and number of rooms.

4. 2.2. Building the ANN Models

The Neural Network Toolbox in MATLAB R2010a was used in constructing the ANN total cost model estimation. The total cost and corresponding parameter data from 30 building projects were divided into three sets: 60% for training the neural network, 20% for validation and 20% for testing. These sets were randomly selected in MATLAB. In this paper, the tan-sigmoid transformation function was applied in the hidden layer and a linear transformation function was used in the output layer, similar. The tan-sigmoid function returns values in the range of -1 and +1. Before training, the data were first normalized to this range to speed up the training process and to improve the network performance. Equation 1 presents the formula used to normalize the training set so that they fall within the range of [-1, +1].

$Y = \frac{Y_{max} - Y_{min}(X - X_{min})}{(X_{max} - X_{min}) + Y_{min}}$………………………………………… (10)

The feed forward back propagation technique was used to generate the best model for the total structural cost. The back propagation algorithm gradually reduces the error between the model output and the target output by minimizing the mean square error (MSE) over a set of training set The MSE is a good overall measure of the success of the training process .The MSE is expressed as (Eq. 2):

$$\mathrm{MSE} = \frac{\sqrt{\sum_{i=1}^{n}(X_i - E(i)2)}}{n} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots..(11)$$

Where: n = number of samples in the training set; xi = model output related to the sample i; E(i) = target output, i.e. total structural cost The weights and bias values are updated according to the Levenberg-Marquardt network training function. This is often the fastest back propagation algorithm and highly recommended, though it requires more memory that other algorithms (Neural Network Toolbox).

4. 2.3. Trial Models

The best ANN model to estimate the total structural cost of buildings was determined by defining the number of neurons (nodes) in the input and output layers, number of hidden layers and the number of neurons in each hidden layer. The model generated utilizes the six input variables. There is no specific rule in determining the number of hidden layers and the number of neurons in each hidden layer. For simplicity of the current problem, one hidden layer was used and the following rules were employed to determine the optimum number of neurons for a network with 14 and 3 input variables.

4. RESULT AND DISCUSSION

4.1 Support vector machines (SVM) predictive model

The Support vector machine (SVM) is a relatively new modelling method that has been rapidly developed in recent years, showing very good results at obtaining accurate models for many problems like pattern classification, function approximation, and regression problems.

Table No 01Estimation of the accuracy for validation data for SVM Model

Coefficient of determination R2	0.95571 (95.571%)
Correlation between actual and predicted target values r	0.997880
MAPE	0.3034884

4.2Predicting construction cost using ANN Model

4. 2.3. Trial Models

The best ANN model to estimate the total structural cost of buildings was determined by defining the number of neurons (nodes) in the input and output layers, number of hidden layers and the number of neurons in each hidden layer. The model generated utilizes the six input variables. There is no specific rule in determining the number of hidden layers and the number of neurons in each hidden layer. For simplicity of the current problem, one hidden layer was used and the following rules were employed to determine the optimum number of neurons for a network with 14 and 3 input variables.

Table No 02Estimation of the accuracy for validation data for ANN Model

Coefficient of determination R2	0.99971 (99.991%)
Correlation between actual and predicted target values r	1.365425
MAPE	0.152541

4.3 Comparison of SVM and ANN

This section presents the best achieved results for MLP models. In table 2 is shown the computed values of R2, MSE, and RMSE for (15) developed ANN models. The network structure is involved three layers and denoted by three numbers, first number indicates the number of neurons in the input layer, second number indicates of the number of neurons in the hidden layer, and third number refers of the neuron in the output layer. (70) Samples are used for training each model, (14) samples are tested of each model. (15) Models are developed.

Artificial Neutrals Network (ANN) approach is used for prediction of construction cost. The selection of transfer functions may strongly impact performance of neural networks. The purpose of this study is to design construction cost prediction model and compare with Regression Analysis and Support Vector Machine considering materials cost as input parameters. We shall stress here the importance of choosing the relevant predictors for the target variable. The actual values of the predictors (real time, contracted price and contracted time), and the target variable (real price) are not used as input variables of the model, but logarithm of their values, which contributed very much for the accuracy of the model. Usually R^2 and MAPE (mean absolute percentage error) are being used as estimators of the accuracy of the predictive models. In statistics, the coefficient of determination R^2 measures the general suitability of the predictive model, indicating how good data points match a curve or a line. The values of R^2 belong to the interval and expressed in percent R^2 computes how much the variation of the output response is attributed to the predictors of the model. The value R^2= 0.095 is for SVM and 0.99 is for ANN model.

Table No 03 Comparison of accuracy for SVM and ANN cost estimation model

Sr. No	Parameter	SVM	ANN
1	Coefficient of determination R^2	0.95571 (95.571%)	0.99971 (99.991%)
2	Correlation between actual and predicted target values r	0.997880	1.365425
3	MAPE	0.3034884	0.152541

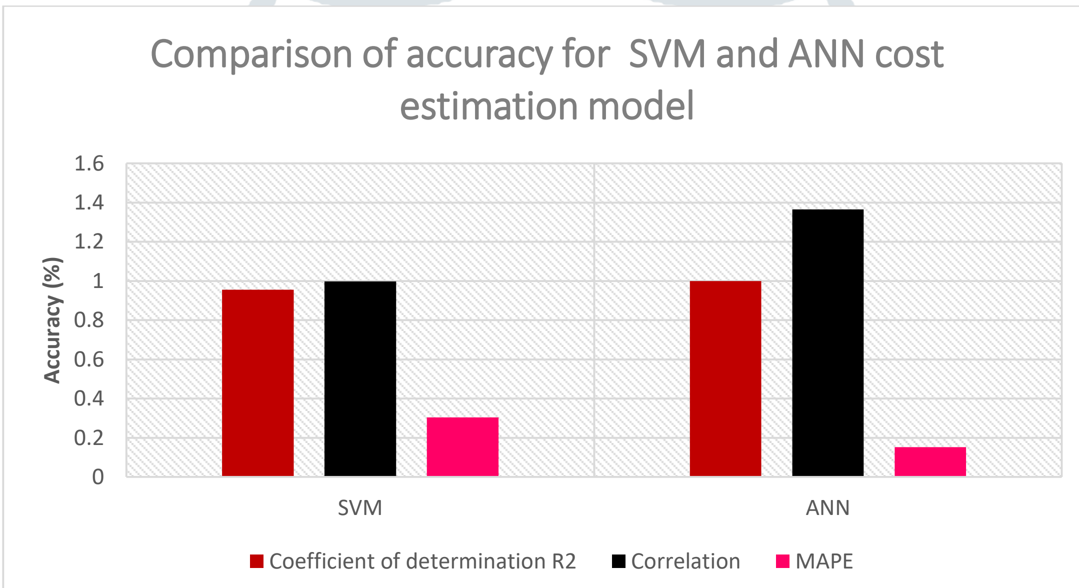

Fig. 05 Comparison of accuracy for SVM and ANN cost estimation model

MAPE usually expresses accuracy as a percentage. For this model MAPE = 0.30 is for SVM and 0.15 is for ANN means that the percentage error of the SVM is high and ANN is low that means ANN is more accurate. The Support Vector Machine model accuracy is R2 = 0.955 i.e (95.5%) for SVM and R2 =0.99 i.e (99.9%) is for ANN. As it can be seen from the results, the SVM model gave excellent predictive result for our data and significantly more accurate in comparison with linear regression. We should stress here that when we do not know in advance the type of the relationship between the predictors and the dependent variable, then we should try several predictive models in order to obtain the best one for our data, which will give most accurate predicting, because for different data, different type of predictive model can be suitable.

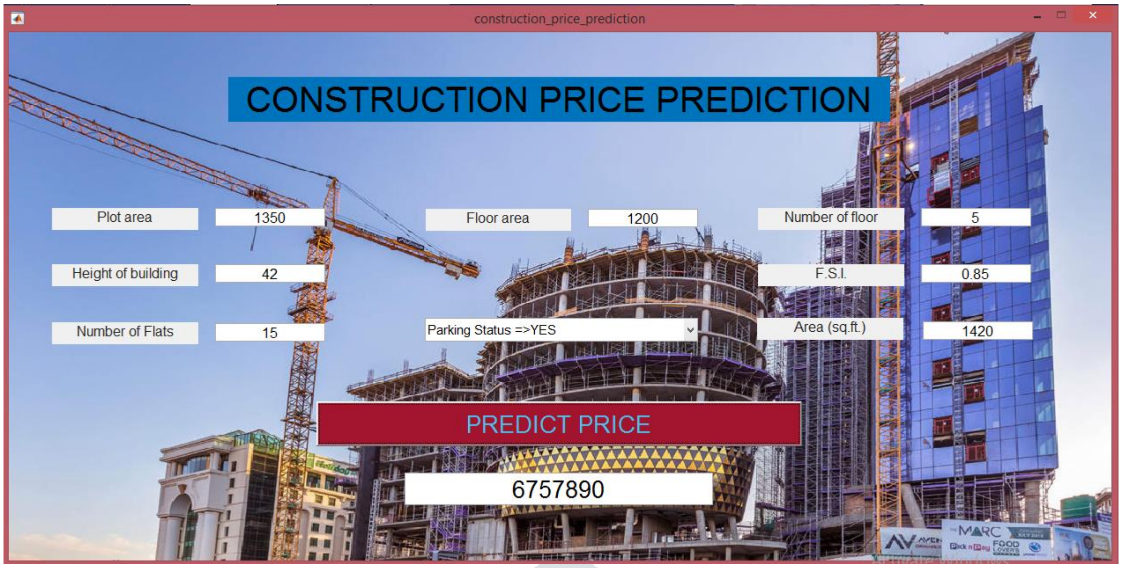

Fig. 06 UI-1 of ANN cost estimation model

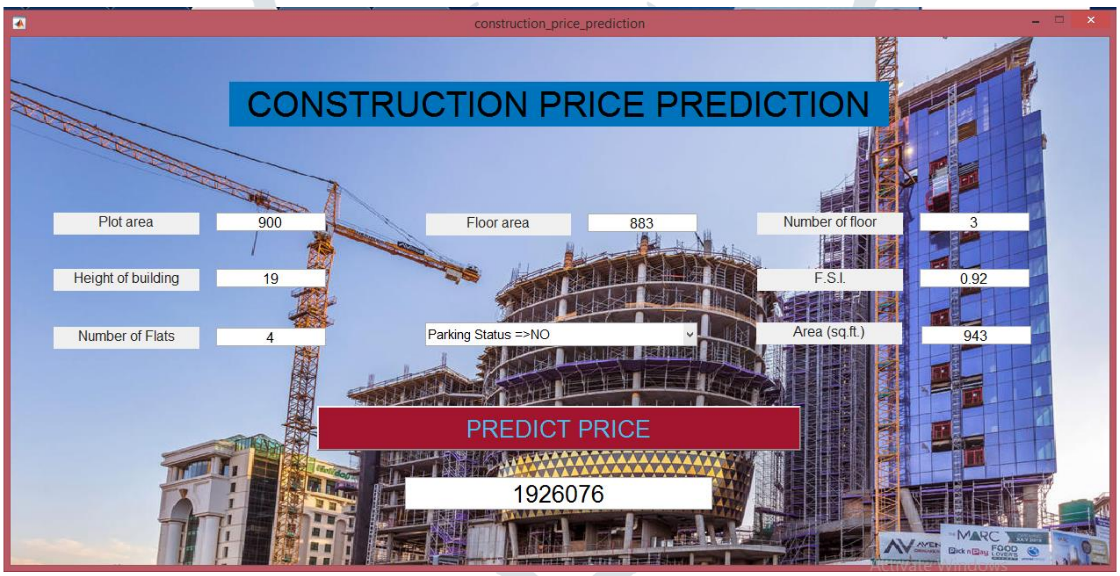

Fig. 07 UI-2 of ANN cost estimation model

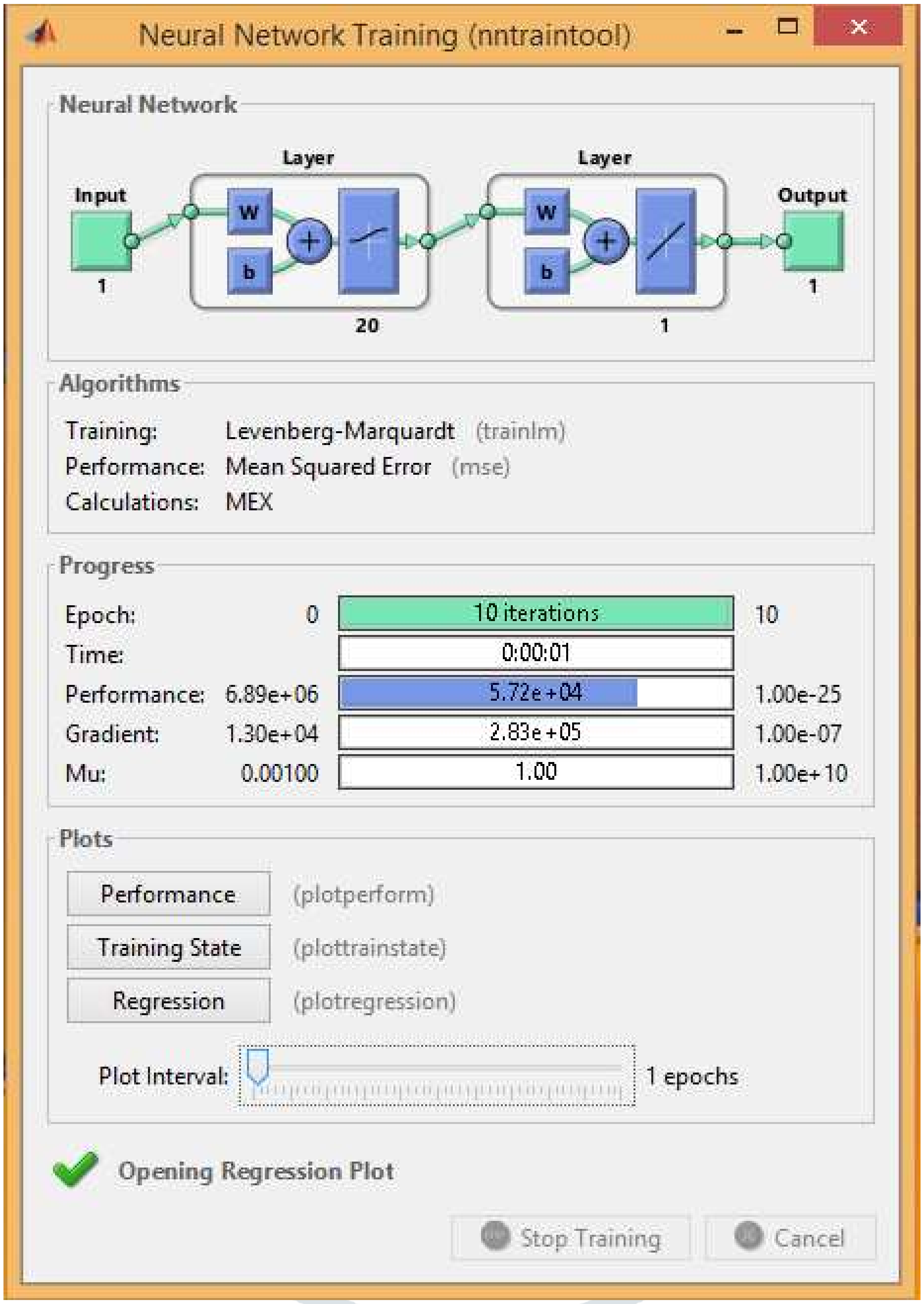

Fig. 08 Training Phase ANN cost estimation model

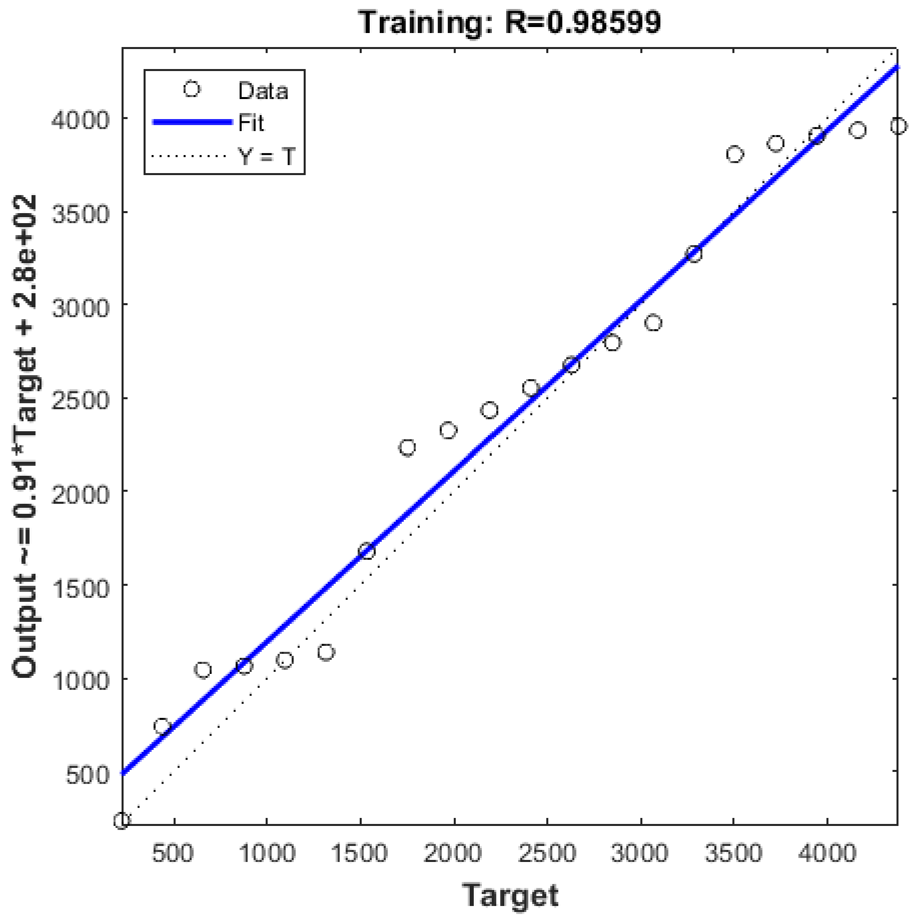

Fig. 9 Training and Target Phase ANN cost estimation model

Figure shows the front end of our system. It contains many field which has to be filled to get your result. One has to click on predict price button to get predicted. Once you click on predict price button the values are feed to the neural network. After that neural network work on that values and reflects the required cost. This feedback cost from neural network is reflected on front end of the system as a predicted value.

CONCLUSION

- In this study investigation, two stages were carried out to achieve the objectives. Data collection and analysis, and developing ANN model have been done.
- This study is deduced in few points as following: Fifteen (15) NNs models were built to predict the cost of the project by using neural network Tool Box software by MATLAB program Through five attributes were taken as predictor variables namely; collect data, preprocessing data, architecture the network, training the model, and testing the model using excel sheet and MATLAB. RMSE, MSE, MAPE, and R2 were calculated.
- The construction process is a complex process that is influenced by numerous and changeable factors. Additionally, the accuracy of construction costs forecasting can have an essential role for the process of construction and for the project participants business. In this paper author compare construction cost using SVM, RA and ANN Therefore, the cost forecasting is a particularly difficult and responsible process.
- Learning from previous projects costs experience is an important issue. For that purpose, a data base for costs of previously realized construction project was formed. MAPE usually expresses accuracy as a percentage. In this project the best achieved results for MLP models. In table 2 is shown the computed values of R2, MSE, and RMSE for (15) developed ANN models. The network structure is involved three layers and denoted by three numbers, first number indicates the number of neurons in the input layer, second number indicates of the number of neurons in the hidden layer, and third number refers of the neuron in the output layer. (70) Samples are used for training each model, (14) samples are tested of each model. (15) Models are developed. MAPE usually

expresses accuracy as a percentage. For this model MAPE = 0.30 is for SVM and 0.15 is for ANN means that the percentage error of the SVM is high and ANN is low that means ANN is more accurate.

- The Support Vector Machine model accuracy is R2 = 0.955 i.e (95.5%) for SVM and R2 =0.99 i.e (99.9%) is for ANN. One of the weaknesses of the SVM model is its speed of convergence, in relation to the linear regression model. The models are convenient for rapid and efficient forecasting of construction cost and they are not a substitution of detail cost estimation process. Due to that, they are applicable for the initial phase of the construction projects by project participants and by clients.
- These models limitation is that they are applicable in construction projects without strong influence of physical factors (poor organization of construction site and works, incomplete documentation, incorrect documentation, bad climate conditions MAPE usually expresses accuracy as a percentage.

FUTURE SCOPE

- This subject opens the door for a lot of future researches.
- The following potential areas of studies, if explored, would provide increased validity to the findings of this thesis: For cost estimation, it is suggested to study all factors separately or as groups and develop the best model, the data collected are not sufficient to cover all government buildings, thus we have to collect more than and add them to the model to improve the error and the missing data must be collected. The model should be augmented to take into consideration the other different types of construction projects. For example: the medical, commercial and other administrative construction projects.
- The model should be applied to predict the duration, productivity, risk analysis, and claims in construction projects.

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